

The Price of Tragedy: Exploring Income and Overdose Deaths in Phoenix

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Introduction:

Overdoses take thousands of lives each year around the world and in the United States. Each year, more and more people die from overdose and the United States continues to have the highest overdose rate year in and year out. Overdosing continues to have a great effect on the people of the United States and their families. In our project, we look at overdoses in one of the largest cities in the US, Phoenix. We look at every zip code in the city of Phoenix and examine the possible correlation between overdose deaths and median household income in every zip code. In doing this, we hope to figure out if the areas of a big city with less financial income tend to die from overdose more or on par with those who are more wealthy.

By conducting this experiment, we can provide truthful strong evidence that can be used to help fight against overdose in Phoenix, and possibly other big cities in the United States that struggle with this problem. The data in this study could show people that an effective way of stopping and limiting the number of overdose deaths in big cities is to help strengthen the communities with lower household income and who worry if their next paycheck is enough so overdose deaths don't continue at a high rate and hurt families and communities.

Methods:

We are utilizing two data sets for analysis: demographic data from Point2Homes and fatal overdoses data from the City of Phoenix open data. The demographic data showing the median household incomes by zip code, population by zip code, and number of homes was created from the U.S. Census Bureau and the 2022 American Community Survey conducted by the Census Bureau. The overdose death data is from the City of Phoenix and was collected by the Phoenix Fire Department and the Maricopa County Office of the Medical Examiner. Both of these data sets are reliable sources of information with how much data they provide and the trustworthy way of obtaining the data. The datasets can be merged on the "zipcode" column in the code, making sure that each zip code is associated with both median income and overdose data.

This experiment uses an observational design to find correlations and patterns by comparing health and socioeconomics in specific areas of Arizona. This experiment can be reproduced in Python by utilizing the two datasets in CSV format and cleaning the data. The data can be cleaned by removing missing data and only using relevant columns of the data, for example, in the household income data, the average household income column is not needed for this experiment. The zip codes need to be formatted consistently across both data sets. Next, the data needs to be merged on the zip code key. Once the data is merged, analyses can be conducted to see correlations, regression, and plots. Our primary method of analysis was finding basic statistical information and finding a correlation analysis. Refer to the appendix to see the full statistical and correlation analyses. We calculated basic statistics for the data, including the mean, median, standard deviation, minimum, and maximum values for both income and overdose deaths. A correlation matrix was computed to assess the relationship between the variables.

In the data, there are a few potential sources of bias. The first is the underrepresentation of overdose deaths. Some overdose deaths may go unreported and be unaccounted for. To allocate for this bias, we can look at some other sources and see how they compare to the City of Phoenix open data source. Another bias that could be present is the potential of some zip codes having more data than others

if they have higher populations or are more focused on by health authorities. We can see the distribution of the data to find any large differences between zip codes.

These findings could be used to help inform local health officers or community organizations to help mitigate overdose crises or see if any zip code has more of a correlation between household income and overdoses. With the acknowledgment of biases, this experiment can provide a deep understanding of the relationship between the two data sets and help inform research and strategies to improve public health.

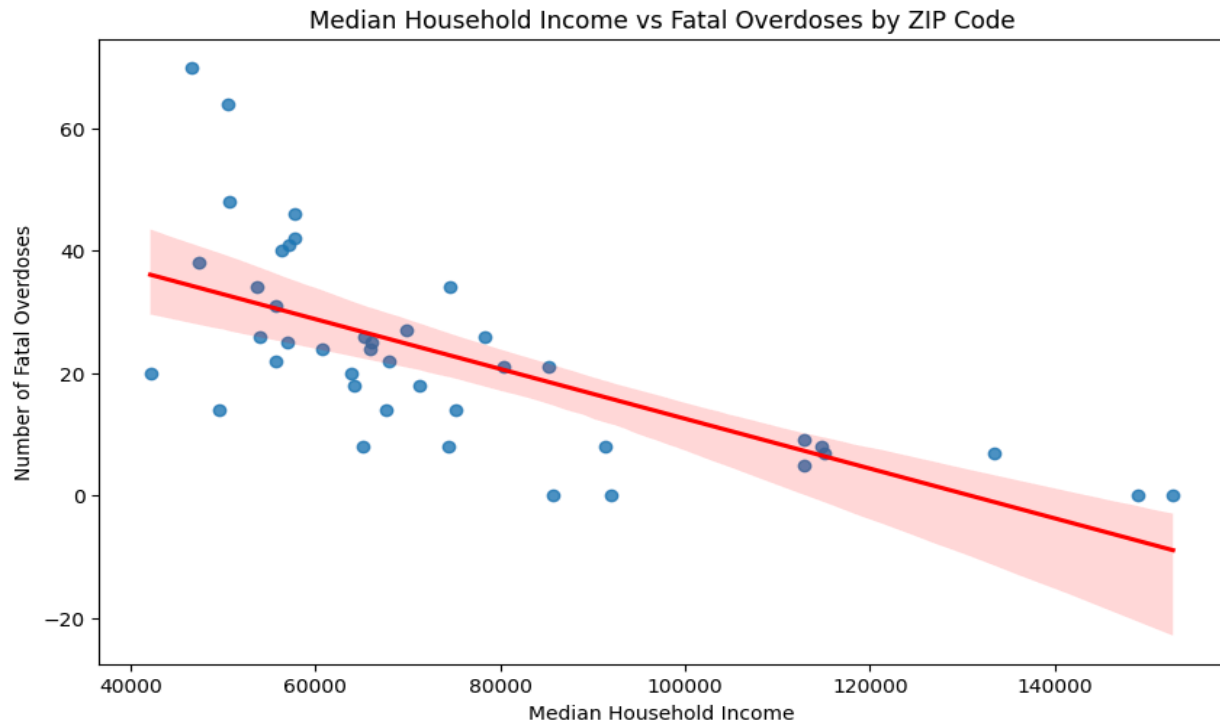
Using Google Collab, Python, knowledge from prior coding classes, and the data we found, we were able to create code that gave us numerous data summaries and some graphs to help interpret the data. The code made it much easier to find statistics on the data, such as a five-number summary of each variable, giving us more information to analyze whether household income affects the number of overdose deaths in an area.

Data Summaries & Visualization:

As shown in point 1 of the Appendix, using code we were able to get a table showing the number of houses, median household income, and overdose deaths for every one of the 42 zip codes we found data for in Phoenix. This helps give a very easy-to-read table for us to begin looking at the correlation and seeing some of the different zip codes where our hypothesis could be correct and some that defy our hypothesis.

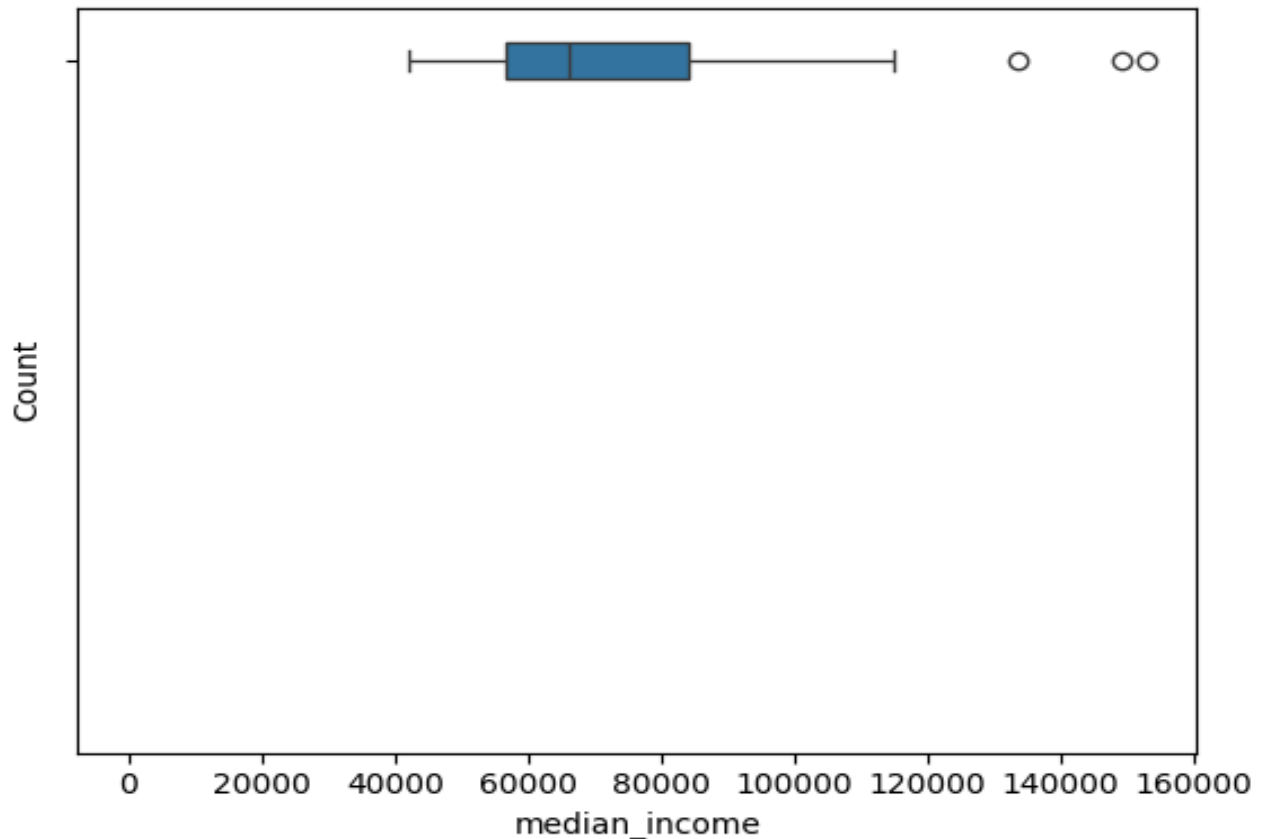
Breaking down our code (can be found in the “References” section) one step at a time, after getting each dataset into its own data frame, we merged the two datasets into `merged_df`. It has 42 rows of data (one row for each zip code). Next, we described the data using the `.describe()` function. The mean value for `number_of_households` is \$34,461, for `median_income` it's \$74,982 and for overdose deaths, it's 22.74. The standard deviation for overdose deaths is 16.355889. This shows that there are variations in the amount of overdose deaths by zip code. The `median_income` values range from \$42,177 to \$152,805, while the overdose death values range from 0 to 70. We found a negative correlation of -0.6785 when looking at median household income versus overdose deaths, which shows that higher median incomes tend to be associated with lower overdose rates, and vice versa.

We visualized the data using scatter plots and box plots. In the scatter plot shown below, we compared median household income and overdose rates.



A downward trend is shown, supporting the negative correlation we found. The regression line is shown in red, and the shaded area represents the confidence interval for the fit. The x-axis represents the median household income (in dollars) for each zip code. The y-axis represents the number of fatal overdoses reported in those zip codes. Each point on the scatter plot represents a specific zip code and the values for income and overdose death rates in that region. Despite the negative correlation, the points are spread out. This could mean that other factors, beyond income, may also contribute to overdose deaths, as we stated earlier when mentioning biases in the data. There are some points far away from the regression line. For example, some zip codes with high median household income have relatively high overdose rates, which shows some of the bias mentioned earlier with other factors contributing to overdose deaths. There is a clear negative correlation shown in the scatterplot, showing that income is a contributing factor to overdose deaths in Arizona. However, the scatterplot only shows a trend, not a direct connection between the median household income and fatal overdoses. We would need further research to see what other factors could contribute to the overdose deaths.

In the box plot shown below, the distribution of median household income across zip codes is shown.



There are three outliers shown on the right, showing that some zip codes with a higher median income have many overdose deaths in their region. The median is a bit skewed to the right, which shows that lower-income households may have more overdose deaths in their zip code.

In conclusion, through statistical analysis, we found that higher median income in certain zip code areas correlated with lower fatal overdose rates. There are definitely other socioeconomic factors that can contribute to the number of overdose deaths, but we did find somewhat of a negative correlation between the two factors. By finding patterns in fatal overdoses across Phoenix, this study shows the importance of economic and geographic factors (median household income, for example) and can be used when designing healthcare programs or creating policies based on different locations in Phoenix. There could be further research done to find other factors contributing to Phoenix overdose levels, but we have found that household income does correlate negatively with fatal overdoses.

Analysis:

Using our notebook and coding, we computed OLS Regression statistics, shown below.

OLS Regression Results

Dep. Variable:	overdoses	R-squared:	0.460
Model:	OLS	Adj. R-squared:	0.447
Method:	Least Squares	F-statistic:	34.12
Date:	Tue, 03 Dec 2024	Prob (F-statistic):	7.90e-07
Time:	18:07:16	Log-Likelihood:	-163.51
No. Observations:	42	AIC:	331.0
Df Residuals:	40	BIC:	334.5
Df Model:	1		

Covariance Type: nonrobust

	coef	std err	t	P> t 	[0.025	0.975]
const	53.2673	5.553	9.592	0.000	42.044	64.491
median_income	-0.0004	6.97e-05	-5.841	0.000	-0.001	-0.000

Omnibus: 7.471 **Durbin-Watson:** 1.717
Prob(Omnibus): 0.024 **Jarque-Bera (JB):** 6.300
Skew: 0.785 **Prob(JB):** 0.0429
Kurtosis: 4.065 **Cond. No.** 2.36e+05

The R-squared statistic at .460 suggests that approximately 46% of the variance for our dependent variable overdoses can be attributed to our independent variable median household income. This shows a moderate fit for our variables and dataset. Adj. R-squared being below our R-squared shows that predictors would not improve our model and results.

Both the F-statistic and Prob(F-Statistic) indicate that the overall model we have created is statistically significant and the outcome could likely be attributed to the relationship between the two variables. The AIC, Log-Likelihood, and BIC are both measures for model comparison. If we were to compare how these variables relate and the strength of our model to another model we could see which model has the lower BIC and AIC value and is therefore the better model.

Moving down to the lowest section of the regression results, we can see that the Omnibus and Prob(Omnibus), along with the Jarque-Bera and prob(JB) for our model tell us that the residuals are not perfectly normally distributed, but they only deviate mildly and is not a great cause for concern. Our Durbin-Watson statistic at 1.717 suggests mild positive autocorrelation which should not lead to a violation of independence between variables or much unreliable results for our model. The skew and kurtosis values for our model indicate moderate skewness in our data which we were able to see through the boxplot we created. Finally, the condition number statistic indicates that one of our variables may be in much larger units than the other, which is correct as median household income is much larger than the number of fatal overdoses for every zip code.

The next analysis we did was a confidence interval code as shown in Appendix 5, and our statistics in Figure 1. For this output, we used a 95% confidence interval and const is the number of fatal

overdoses by zip code. Using this output, we were able to determine that for every 1\$ increase in median income, the number of fatal overdoses is expected to decrease by a value between .000266 and .000548, which supports our linear regression data as well. Using all the data collected and creating new code, we were able to perform a t-test as shown in Appendix 6. Our null hypothesis was: There is no relationship between median household income and fatal overdoses by Arizona zip code, while our alternative hypothesis was: There is a relationship between median household income and fatal overdoses by Arizona zip code. The code gave us a t value of -4.739 and a degree of freedom value of 34.359. Since the t-stat is negative, the mean number of fatal overdoses in the high-income group is lower than in the low-income group. Using these two values we found a p-value of .000018, which allows us to reject the null hypothesis at our .05 significance level. This means there is statistically significant data to show there is a relationship between the median household income and the number of fatal overdoses by zip code in Phoenix.

The final bit of analysis we did was coding to find the needed sample size for 80% power and a power test as shown in Appendix 7 and 8. We found that we'd need about 64 zip codes to have 80% power and a significance level of 0.05. We have only had 42 zip codes in our study which could show bias and increase the risk of a type 2 error. For the power test we conducted with the 42 zip codes, we reached a power of about 62% at a .05 significance level, which is not ideal but there is nothing we could have done as that was all the zip codes on our CSV file.

Conclusion:

In conclusion, this study explored the relationship between median household incomes and fatal overdoses in the 42 zip codes of Phoenix. We were able to find patterns and form conclusions by merging the data and performing analyses to inform public health officials. With statistical analysis, we found a relatively strong negative correlation between the two factors (-0.678). This proved that median household income does have a significant relationship with overdoses whereas income decreases, overdoses increase, and vice versa. This shows that economic factors may play a part in fatal overdoses or other substance use issues in Phoenix.

Using our findings, health professionals and policymakers can find regions in Phoenix where higher overdose rates correspond to lower incomes and allocate resources to help the situation. This data is valuable because it shows the cross-over between different socio-economic lifestyles and what effect it can have on health. By location-specific zip codes where there may be a stronger connection between income and overdoses, policymakers or organizations can help implement programs or resources where they are most needed. For example, the zip codes with higher overdoses and lower median household income can be provided with increased healthcare access or various support services.

This study also shows limitations and biases because the data we used only captures a few pieces of information. There are many other data sources that could be acquired to deepen the research and find new conclusions, such as information about age, race, ethnicity, or the availability of healthcare resources. The biases in the data, such as the potential underreporting of overdoses or the slight unfairness that higher versus lower populated zip codes causes, could limit the information in the conclusions we find.

There are many ways that future researchers could expand on our data and predictions. For example, they could incorporate more and new variables to think about qualitative data to represent the experiences of communities. Also, researchers could look at changes over time in income levels and community support in the issue to see if those have influenced overdose rates. Other factors such as

education levels, unemployment rates, healthcare, and the amount of community support can be considered in future research to provide more insight into the issue.

The conclusions we have drawn from this data have found that there is a dire need to address economic inequalities as a part of a bigger cause of overdoses. Policymakers and health professionals can fund preventative resources for lower-income communities in Phoenix to reduce health inequalities and potentially save lives from overdoses. This research and project shows the important role that economic factors play in the number of overdose deaths in Phoenix, which offers a foundation for future research and potential solutions.

Appendix:

1. Merge datasets & data table (head)

```
merged_df = pd.merge(median_income_df, overdoses_df, on="zipcode")
print(merged_df)
```

	zipcode	number_of_households	median_income	overdoses
0	85003	9085	49607.0	14
1	85004	10548	67727.0	14
2	85006	24308	54073.0	26
3	85007	13939	50490.0	64
4	85008	61759	57728.0	42
5	85009	52420	46707.0	70

2. Code for description of data and boxplot:

```
print(merged_df.describe())
correlation_matrix = merged_df.corr()
print(correlation_matrix)
sns.boxplot(x="median_income", data=merged_df)
sns.histplot(merged_df["overdoses"])
```

	zipcode	number_of_households	median_income	overdoses
count	42.000000	42.000000	42.000000	42.000000
mean	85031.476190	34461.690476	74982.357143	22.738095
std	20.783716	16564.854061	27255.834248	16.355889
min	85003.000000	4845.000000	42177.000000	0.000000
25%	85016.250000	24419.500000	56568.750000	8.250000
50%	85028.500000	34618.500000	66045.500000	21.500000
75%	85042.750000	43793.000000	84080.000000	30.000000
max	85086.000000	68805.000000	152805.000000	70.000000

3. Code for correlation:

```
correlation = merged_df["median_income"].corr(merged_df["overdoses"])\
print(f"Correlation between median income and overdoses: {correlation}")
Correlation between median income and overdoses: -0.6784873358684328
```

4. Code for OLS regression results:

```
import statsmodels.api as sm
from statsmodels.stats.power import TTestIndPower
X = sm.add_constant(merged_df['median_income'])
```

```
y = merged_df['overdoses']
model = sm.OLS(y, X).fit()
model.summary()
```

5. Code for confidence interval:

```
model.conf_int(alpha=0.05) #Confidence interval of 95%
```

6. Code for T-Test:

```
median = merged_df['median_income'].median()
high_income = merged_df[merged_df['median_income'] > median]['overdoses']
low_income = merged_df[merged_df['median_income'] <= median]['overdoses']
mean_high = high_income.mean()
mean_low = low_income.mean()
```

#Standard Deviations

```
std_high = high_income.std()
std_low = low_income.std()
```

```
n_high = len(high_income)
n_low = len(low_income)
```

#T-stats

```
pooled_std = ((std_high ** 2) / n_high + (std_low ** 2) / n_low) ** 0.5
t_stat = (mean_high - mean_low) / pooled_std
```

#Degrees of Freedom

```
df = ((std_high ** 2 / n_high + std_low ** 2 / n_low) ** 2) / \
      (((std_high ** 2 / n_high) ** 2) / (n_high - 1) + ((std_low ** 2 /
n_low) ** 2) / (n_low - 1))
t_stat, df
(-4.739194839856897, 34.35864792729901)
```

7. Code for sample size to power:

```
analysis = TTestIndPower()
effect_size = 0.5
alpha = 0.05
power = 0.8
sample_size = analysis.solve_power(effect_size=effect_size, alpha=alpha,
power=power)
```

```
sample_size
63.7656117754095
```

8. Code for power test:

```
effect_size = 0.5
alpha = 0.05
power = 0.62
sample_size = analysis.solve_power(effect_size=effect_size, alpha=alpha,
power=power)
Sample_size
42.03699214498879
```

9. Figure 1:

	0	1
const	42.043803	64.490799
median_income	-0.000548	-0.000266

References:

Mizrahi, A., & Whiting, J. (Year). *Code for analyzing income and overdose deaths in Phoenix*. Google Colab.

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